II. Fill in the blanks.

1. The shoot system grows upward in response to _____
2. _____ is positively hydrotropic as well as positively geotropic.
3. The green pigment present in the plant is _____
4. The solar tracking of sunflower in accordance with the path of sun is due to _____.
5. The response of a plant part towards gravity is _____.
6. Plants take in carbondioxide for photosynthesis but need _____ for their living.

III. Match column A with column B.

Column A	Column B
Roots growing downwards into soil.	Positive phototropism
Shoots growing towards the light.	Negative geotropism
Shoots growing upward.	Negative phototropism
Roots growing downwards away from light.	Positive geotropism

IV. State whether true or false. If false, correct the statement.

1. The response of a part of plant to the chemical stimulus is called phototropism.
2. Shoot is positively phototropic and negatively geotropic.
3. When the weather is hot, water evaporates lesser which is due to opening of stomata.
4. Photosynthesis produces glucose and carbon dioxide.
5. Photosynthesis is important in releasing oxygen to keep the atmosphere in balance.
6. Plants lose water when the stomata on leaves are closed.

V. Answer very briefly.

1. What is nastic movement?
2. Name the plant part
   a) Which bends in the direction of gravity but away from the light.
   b) Which bends towards light but away from the force of gravity.
3. Differentiate phototropism from photonasty.
4. Photosynthesis converts energy X into energy Y.
   a) What are X and Y?
   b) Green plants are autotrophic in their mode of nutrition. Why?
5. Define transpiration.
6. Name the cell that surrounds the stoma.

VI. Answer briefly.

1. Give the technical terms for the following:
   a) Growth dependent movement in plants.
   b) Growth independent movement in plants.
2. Explain the movement seen in Pneumatophores of Avicennia.
3. Fill in the blanks:

$$6CO_2 + ___ \xrightarrow[\text{Chlorophyll}]{\text{Sunlight}} ___ + 6O_2\uparrow$$

4. What is chlorophyll?
5. Name the part of plant which shows positive geotropism. Why?
6. What is the difference between movement of flower in sunflower plant and closing of the leaves in the *Mimosa pudica?*

IX_SCI_EM_Unit-19_BIO.indd 231 30/11/2022 20:12:43